1 Task 1: Stride Pattern Prefetcher (SPP)

1.1 Design overview

SPP is an L1D prefetcher that works per instruction pointer (IP) and never crosses a page boundary. It has three parts.

- **IP table** (64 entries, direct-mapped). Each entry stores the IP tag, the last accessed cache line (12 bits), the last stride, a 2-bit uniform-stride confidence counter and an 8-bit stride-history signature. The index is `(ip ^ (ip>>6)) & 63` and the tag is `(ip>>6) & 0xFF`.
- **Pattern table** (256 entries, indexed by the signature). Each entry holds the predicted next stride and a 2-bit confidence. The signature is updated as `sig' = ((sig << 3) ^ (stride & 0x7F)) & 0xFF`, so it summarises the last few strides of that IP and maps to the stride expected next. A repeating sequence such as $(+1, +2, +2, +2, +3)^n$ therefore trains a chain of table entries.
- **Accuracy checker**. Two small counters (prefetches issued and useful prefetches) are evaluated every 128 prefetches and set the prefetch degree.

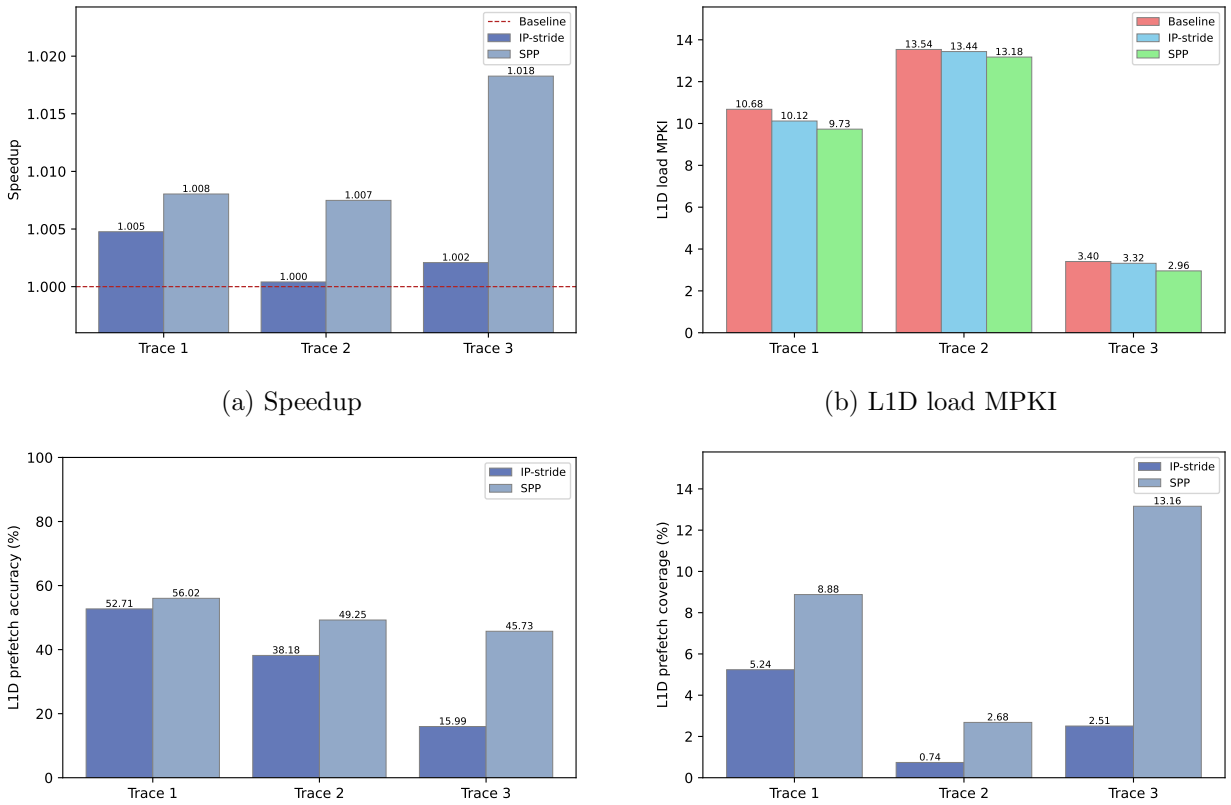
Uniform first, pattern as fall-back. If the uniform confidence of an IP is saturated (value 3), SPP prefetches `addr + k*stride` for $k = 1..$degree. When a stride differs from the previous one the counter is halved, so a broken uniform pattern drops out of the uniform path quickly, and the IP then uses the pattern table. In that mode SPP reads the predicted stride for the current signature, issues the prefetch, advances a speculative signature and repeats up to the current degree. It stops when the pattern-table confidence is below 2, the predicted stride is 0, or the next line would leave the page. Only demand accesses train the tables, and strides larger than 63 lines only resynchronise the IP entry. Prefetches are filled into L1D; if the L1D MSHR is more than half full the prefetch is dropped.

1.2 Results

Speedup is measured against the no-prefetching baseline of the same trace. L1D load MPKI and accuracy are taken from the ChampSim statistics (accuracy is useful prefetches divided by prefetches issued to the lower level), and coverage is the reduction in L1D load misses relative to the baseline.

Trace	Config	IPC	Speedup	L1D load MPKI	Accuracy (%)	Coverage (%)
Trace 1	Baseline	0.232345	1.0000	10.678	–	–
Trace 1	IP-stride	0.233452	1.0048	10.119	52.71	5.24
Trace 1	SPP	0.234213	1.0080	9.730	56.02	8.88
Trace 2	Baseline	0.251158	1.0000	13.538	–	–
Trace 2	IP-stride	0.251260	1.0004	13.438	38.18	0.74
Trace 2	SPP	0.253038	1.0075	13.175	49.25	2.68
Trace 3	Baseline	0.382065	1.0000	3.403	–	–
Trace 3	IP-stride	0.382862	1.0021	3.318	15.99	2.51
Trace 3	SPP	0.389045	1.0183	2.956	45.73	13.16

Table 1: IPC, speedup, L1D load MPKI, prefetch accuracy and coverage for the baseline, IP-stride and SPP.



(a) Speedup

(b) L1D load MPKI

(c) L1D prefetch accuracy

(d) L1D prefetch coverage

Figure 1: SPP compared with the IP-stride prefetcher and the baseline on the three traces.

Discussion. SPP gives a higher speedup than IP-stride on all three traces: 1.0080/1.0075/1.0183 against 1.0048/1.0004/1.0021, a geometric mean of 1.0113 against 1.0024. It also has higher coverage on every trace (8.88%, 2.68%, 13.16% against 5.24%, 0.74%, 2.51%) and higher accuracy. The pattern-table path is the reason. In all three traces most triggers come from non-uniform patterns: 330,222 pattern triggers against 113,216 uniform ones in Trace 1, 185,947 against 43,377 in Trace 2, and 337,323 against 47,749 in Trace 3. A constant-stride prefetcher cannot cover these accesses. Trace 3 benefits most: IP-stride has an accuracy of only 15.99% there, while SPP reaches 45.73% and a 13.16% coverage. The absolute gains are small because the traces are memory bound (baseline IPC between 0.23 and 0.38) and a large share of their load misses follow no stride pattern, so an in-page L1D stride prefetcher can cover only part of the miss stream. The ceiling is set by the traces and not by a single parameter, as the parameter study in Section 1.6 shows.

1.3 Storage overhead

Structure	Field	Bits/entry	Entries	Total bits
IP table	valid	1		
	tag	8		
	last cache line (12-bit)	12		
	last stride (signed)	7		
	uniform confidence	2		
	stride signature	8		
	entry total	**38**	64	**2432**
Pattern table	predicted stride (signed)	7		
	confidence	2		
	entry total	**9**	256	**2304**
Accuracy checker	issued counter (window 128)	8	1	8
	useful counter	8	1	8
	degree register	3	1	3
	total			**19**
Total				**4755 bits (594.4 B)**

Table 2: Bit-level storage of SPP. The pattern table is indexed directly by the 8-bit signature, so it needs no tag.

The issued counter counts prefetches sent to the lower level (an L1D miss) and the useful counter counts demand hits on prefetched L1D lines. Both are cleared at the end of every window of 128 issued prefetches, so an 8-bit counter is enough (the useful counter saturates).

1.4 Accuracy threshold for throttling

The accuracy of the last window (useful/issued) selects the degree: $\geq 75\% \rightarrow 6$, $\geq 50\% \rightarrow 4$, $\geq 25\% \rightarrow 2$, otherwise 1. The bands are evenly spaced quartiles. Below 25% fewer than one prefetch in four is useful, so extra prefetches mostly pollute L1D and use up bandwidth. Above 75% nearly every prefetch is used, so running further ahead is safe. The 50% boundary marks the point where useful prefetches outnumber useless ones. We compared this choice with a stricter set (90/70/40%) on Trace 1 (Section 1.6). The stricter thresholds lowered the degree more often, giving a coverage of 8.03% instead of 8.88% with no gain in IPC, so we kept 75/50/25%. The per-window accuracy varies widely between phases and traces (20.9% at the end of Trace 2, 28.1% for Trace 1 and 85.4% for Trace 3), and the degree histogram in Table 3 shows that all four degree levels are used, so every band of the throttle is exercised.

An earlier version counted every accepted prefetch call as "issued". That included requests that hit in L1D or merged in the prefetch queue and never reach the lower level, so the measured accuracy was far lower than the true accuracy and the degree stayed at 1 (Trace 1: 609,179 of 687,901 accesses at degree 1). Counting only prefetches that are sent to the lower level fixed this.

Trace	Windows	Last-window acc.	Final degree	d1	d2	d4	d6
Trace 1	920	28.1%	2	23,455	310,364	246,089	108,120
Trace 2	464	20.9%	1	124,257	72,756	74,853	131,208
Trace 3	601	85.4%	6	214,524	132,339	71,774	137,795

Table 3: Accuracy-checker behaviour: windows evaluated, accuracy of the last window, final degree, and the number of trained accesses spent at each degree.

1.5 Choice of prefetch degrees

- **Degree 6 ($\geq$ 75%).** Prefetches are rarely wasted, so SPP can run further ahead of the demand stream. This increases coverage and gives late prefetches more time to arrive.
- **Degree 4 ($\geq$ 50%).** Half of the prefetches pay off, so SPP keeps a moderate lookahead and limits pollution.
- **Degree 2 ($\geq$ 25%).** Most prefetches are useless, so SPP keeps only a short lookahead while some coverage remains.
- **Degree 1 (below 25%).** The prefetcher stays active so the tables keep learning and accuracy can recover when the access pattern changes, but pollution is minimal. Degrees 3 and 5 are not used, because the coarse steps make the throttle react clearly to a change of accuracy.

The chain depth is also limited by the pattern-table confidence and the page boundary, so the effective degree on a pattern-driven IP can be lower than the selected degree.

1.6 Parameter study on Trace 1

Two choices were tested on Trace 1 only: the accuracy thresholds, and whether prefetches that find the L1D MSHR busy are pushed to L2 or dropped. The IPC differences between the three variants are below 0.005%, so IPC does not separate them; coverage and accuracy do.

Variant	IPC	Speedup	MPKI	Cov. (%)	Acc. (%)
Initial (issued counter incl. L1 hits, window 256)	0.234072	1.0074	9.954	6.78	57.14
75/50/25, busy MSHR pushed to L2	0.234219	1.0081	9.736	8.82	54.52
90/70/40, busy MSHR pushed to L2	0.234208	1.0080	9.821	8.03	56.51
75/50/25, busy MSHR dropped **(final)**	0.234213	1.0080	9.730	8.88	56.02

Table 4: Design variants on Trace 1.

We chose 75/50/25 with dropped prefetches. It has the best coverage and lower load MPKI than the other threshold set, with essentially the same IPC, and it removes the path to L2.